
The Vicsek Model

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What is Vicsek Model?

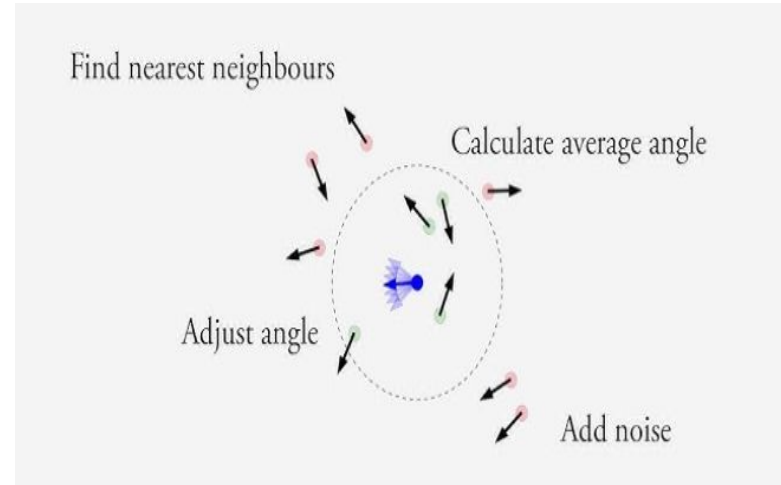
A system of self-propelled particles is evolved in time using only two simple rules:

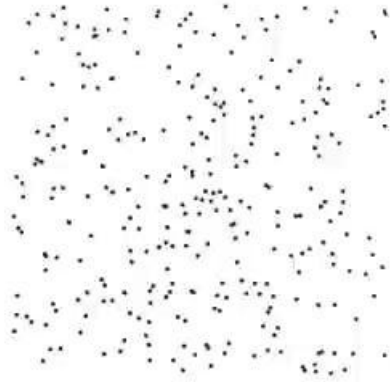
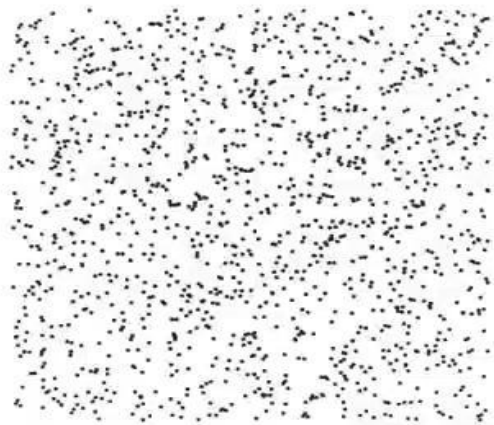
$$\mathbf{x}_i(t + \Delta t) = \mathbf{x}_i(t) + \mathbf{v}_i(t)\Delta t$$

$$\theta(t + \Delta t) = \langle \theta(t) \rangle_r + \Delta \theta$$

$\swarrow$ $\searrow$

$$\arctan \left(\frac{\langle \sin(\theta(t)) \rangle_r}{\langle \cos(\theta(t)) \rangle_r} \right) \quad \left[-\frac{\eta}{2}, \frac{\eta}{2} \right]$$





	ρ	L	η
First	6.12	15	2.0
Second	0.48	25	0.1
Third	12	15	0.2



High density and noise, particles move randomly with some correlation.



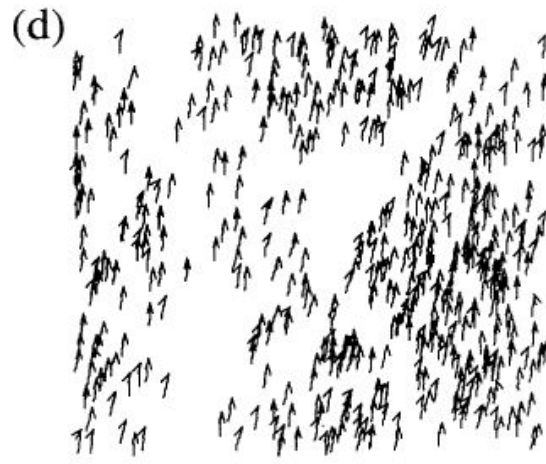
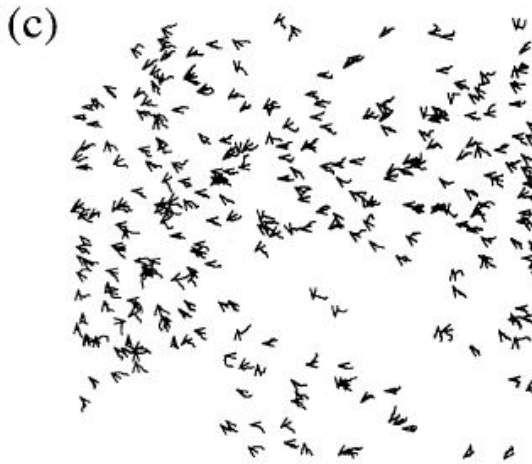
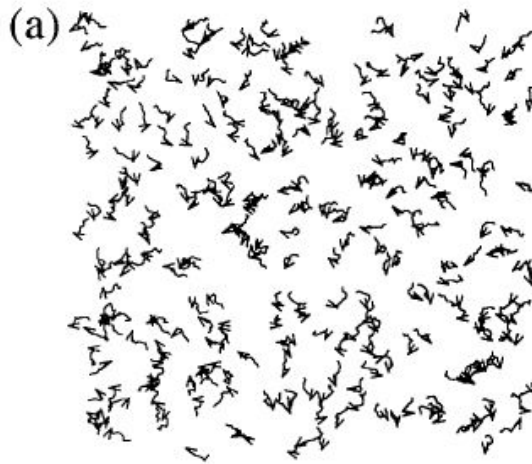
Small density and noise, particles tend to form groups in random directions



High density and small noise, order motion.

Alignment

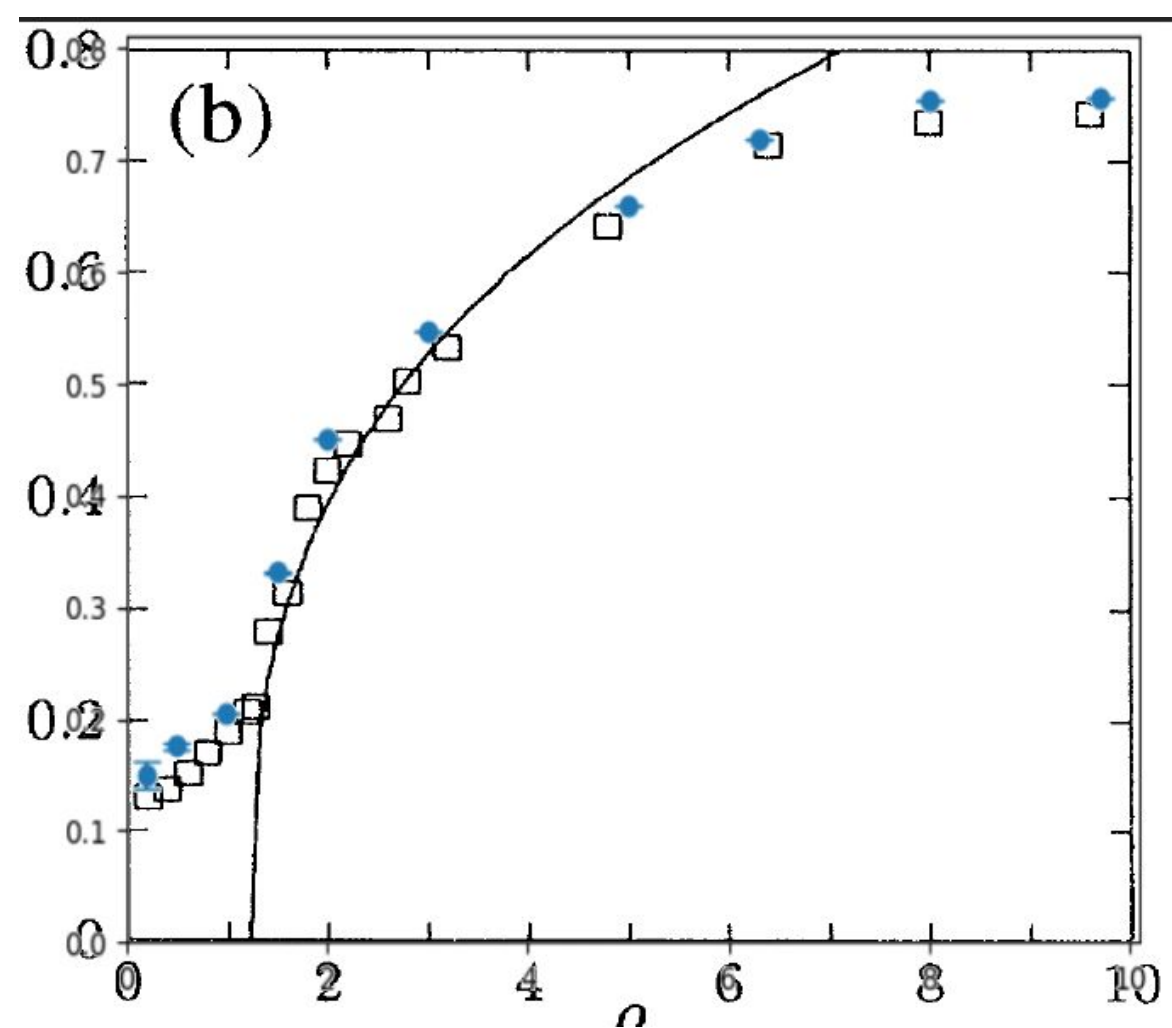
$$v_a = \frac{1}{N_v} \left| \sum_{i=1}^N v_i \right|$$



Alignment vs. density

Low density $\longrightarrow$ weak alignment:

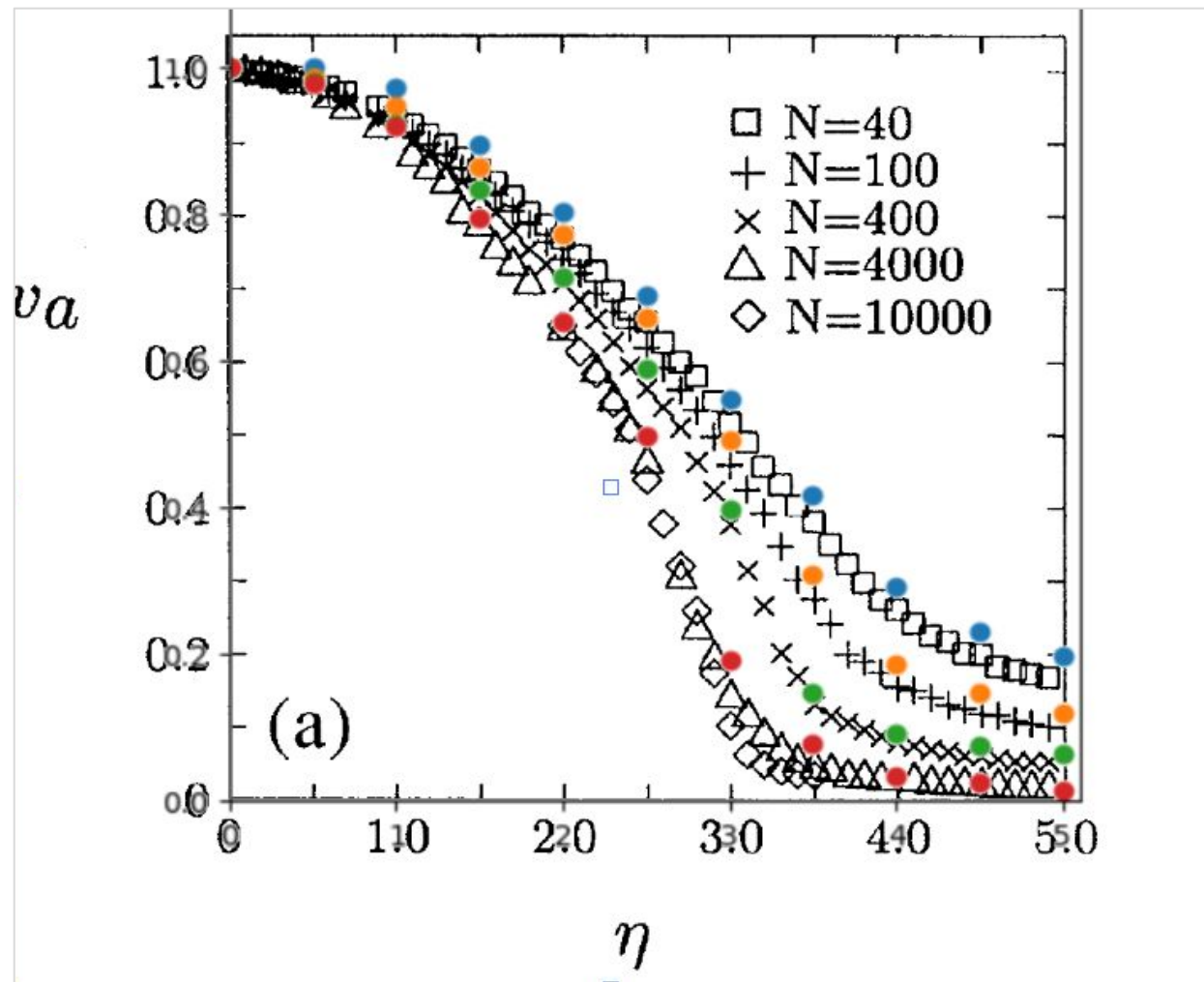
- Each particle interacts with few neighbors
- Direction updates are dominated by noise



Alignment vs. Noise

At fixed density:

$$L_{40} < L_{100} < L_{400} < L_{4000}$$



What if we bring up leaders?

Leaders → weighted particles

Followers:

$$\theta_i(t + \Delta t) = \langle \theta_j(t) \rangle_{j \in \mathcal{N}_i \setminus \{leader\}} + w \theta_{leader}(t) + \xi_i(t)$$

How much a leader affect followers

Leader:

$$\theta_i(t + \Delta t) = \frac{1}{w} \langle \theta_j(t) \rangle_{j \in \mathcal{N}_i \setminus \{i\}} + \theta_i(t) + \beta \xi_i(t)$$

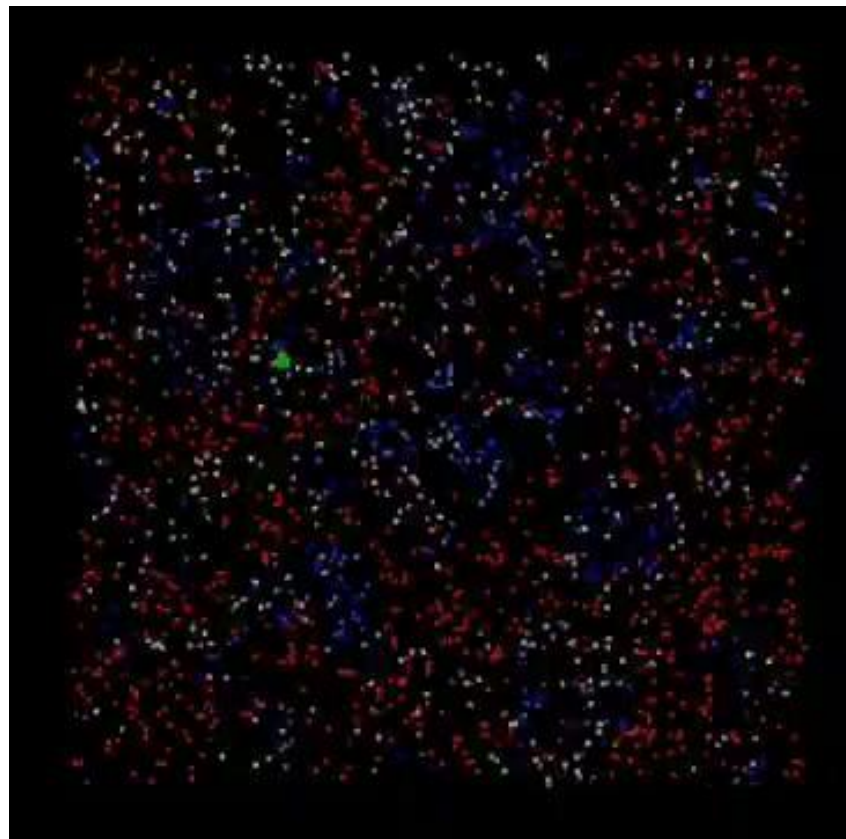
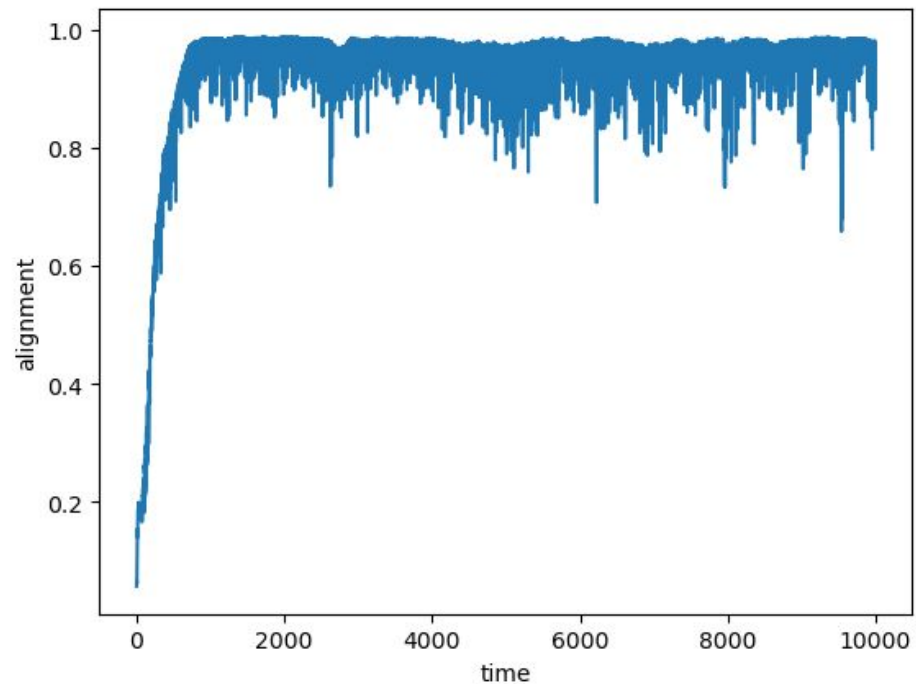
How much a follower affect a leader

Leader noise strength controller

Alignment:

$$A(t) = \langle \cos(\theta_k(t) - \theta_L(t)) \rangle_{N_f} = \frac{1}{N_f} \sum_{k \in \mathcal{F}} \frac{\mathbf{v}_k(t) \cdot \hat{\mathbf{n}}_L(t)}{v_0}, \quad \hat{\mathbf{n}}_L(t) = \frac{\mathbf{v}_L(t)}{v_0}.$$

Alignment vs. Time



$\eta = 0.5$, $w = 2.0$

Blue $\rightarrow 1$

White $\rightarrow 0$

Red $\rightarrow -1$

Mean Alignment vs. weight

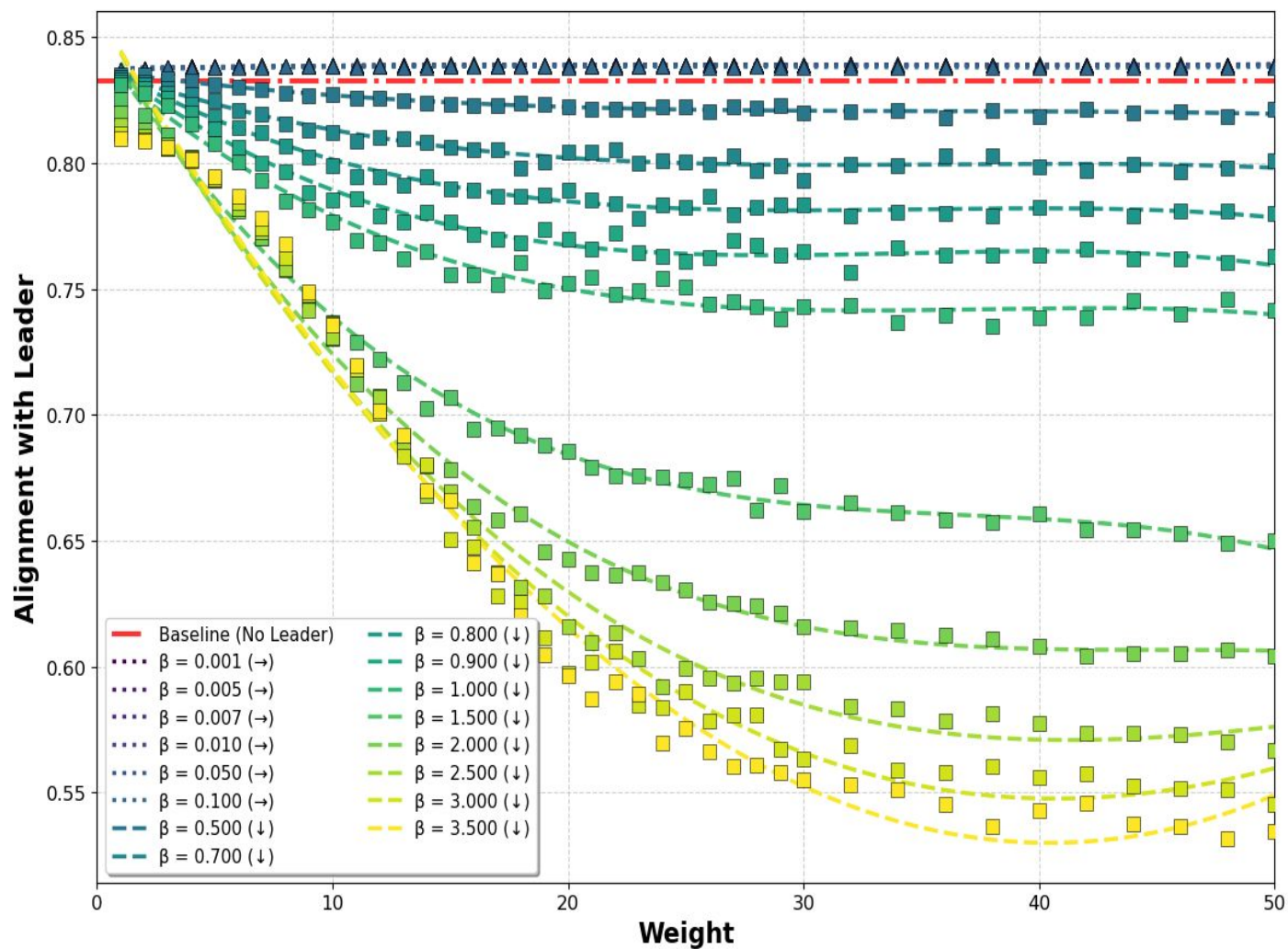
$$\theta_i(t + \Delta t) = \frac{1}{w} \langle \theta_j(t) \rangle_{j \in \mathcal{N}_i \setminus \{i\}} + \theta_i(t) + \beta \xi_i(t)$$

As weight increase:

1. Followers affect the leader less
2. Motion dominates by its own dynamics and noise
3. Followers increasingly attempt to follow a noisy and incoherent leader

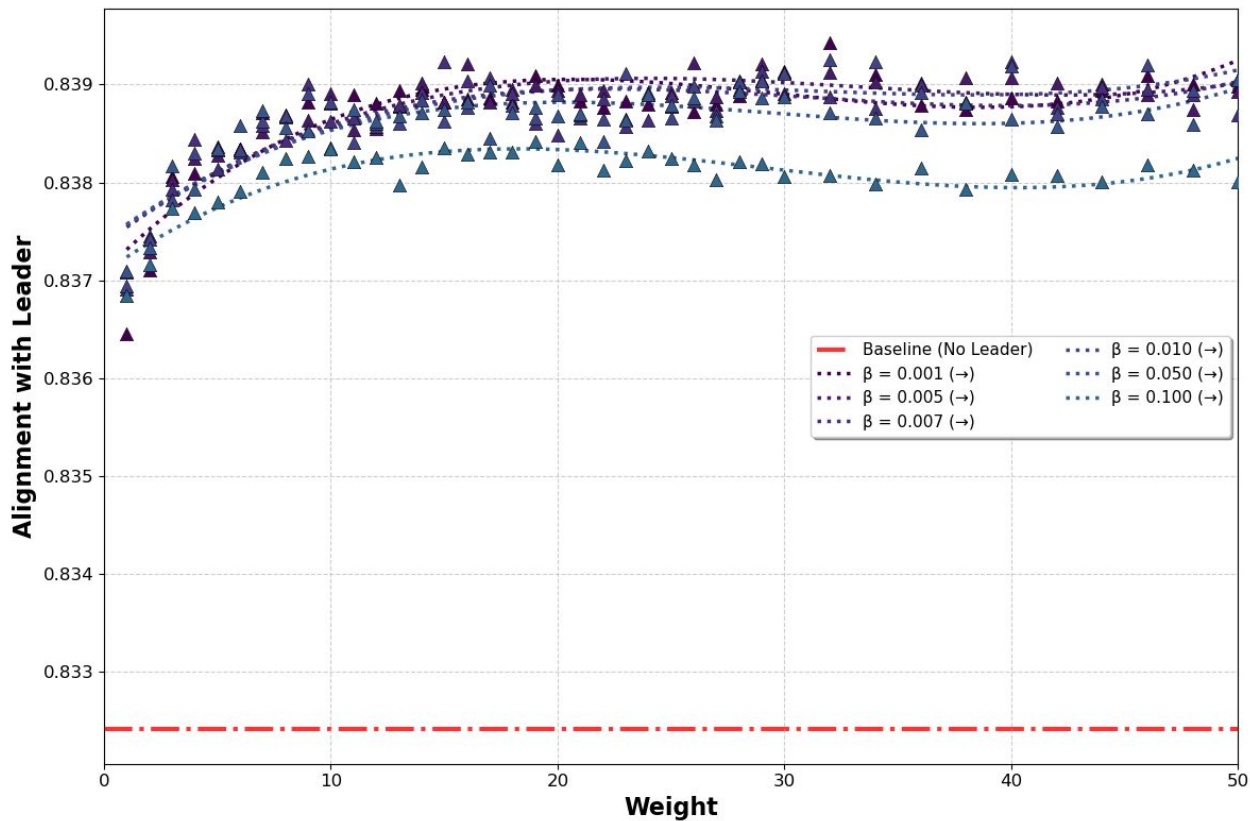


The followers end up tracking a noisy reference, which reduces the global alignment.



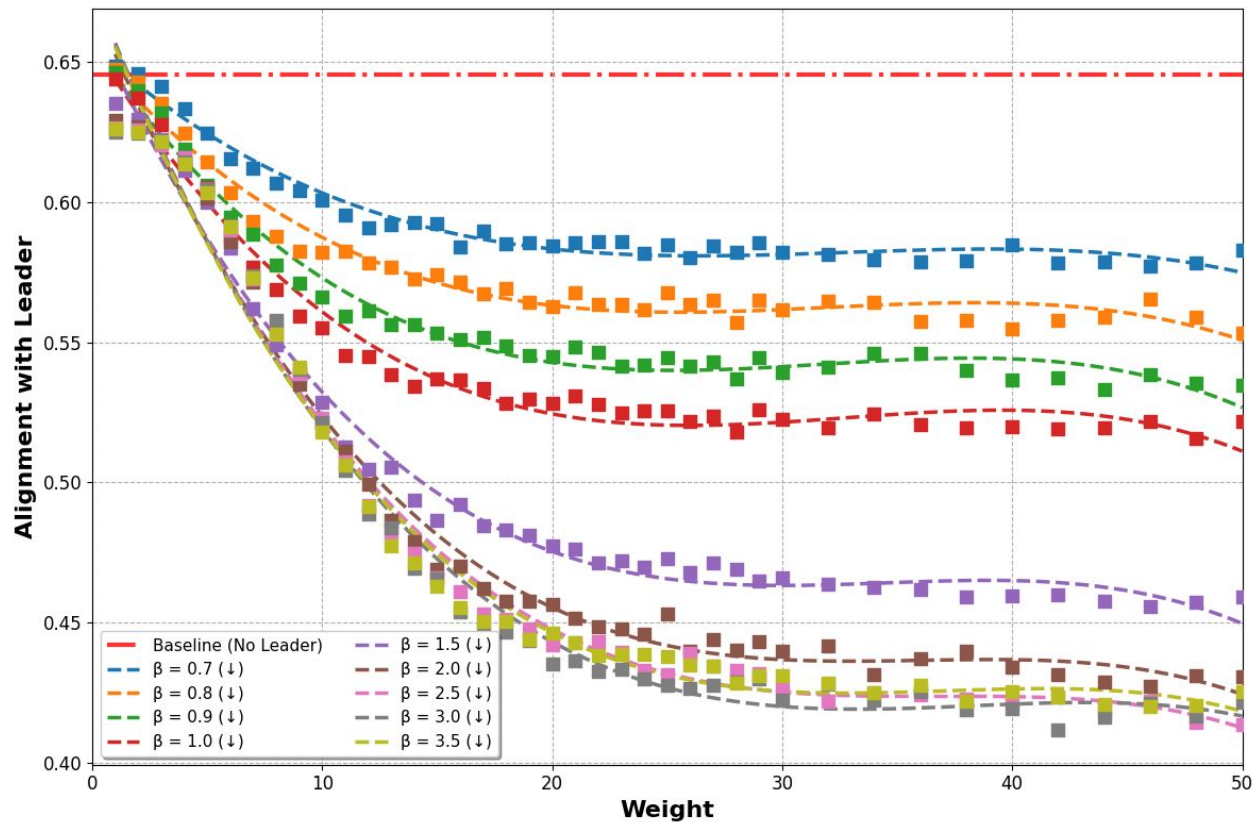
Above Baseline

$$\theta_i(t + \Delta t) = \frac{1}{w} \langle \theta_j(t) \rangle_{j \in \mathcal{N}_i \setminus \{i\}} + \theta_i(t) + \beta \xi_i(t)$$



Below Baseline

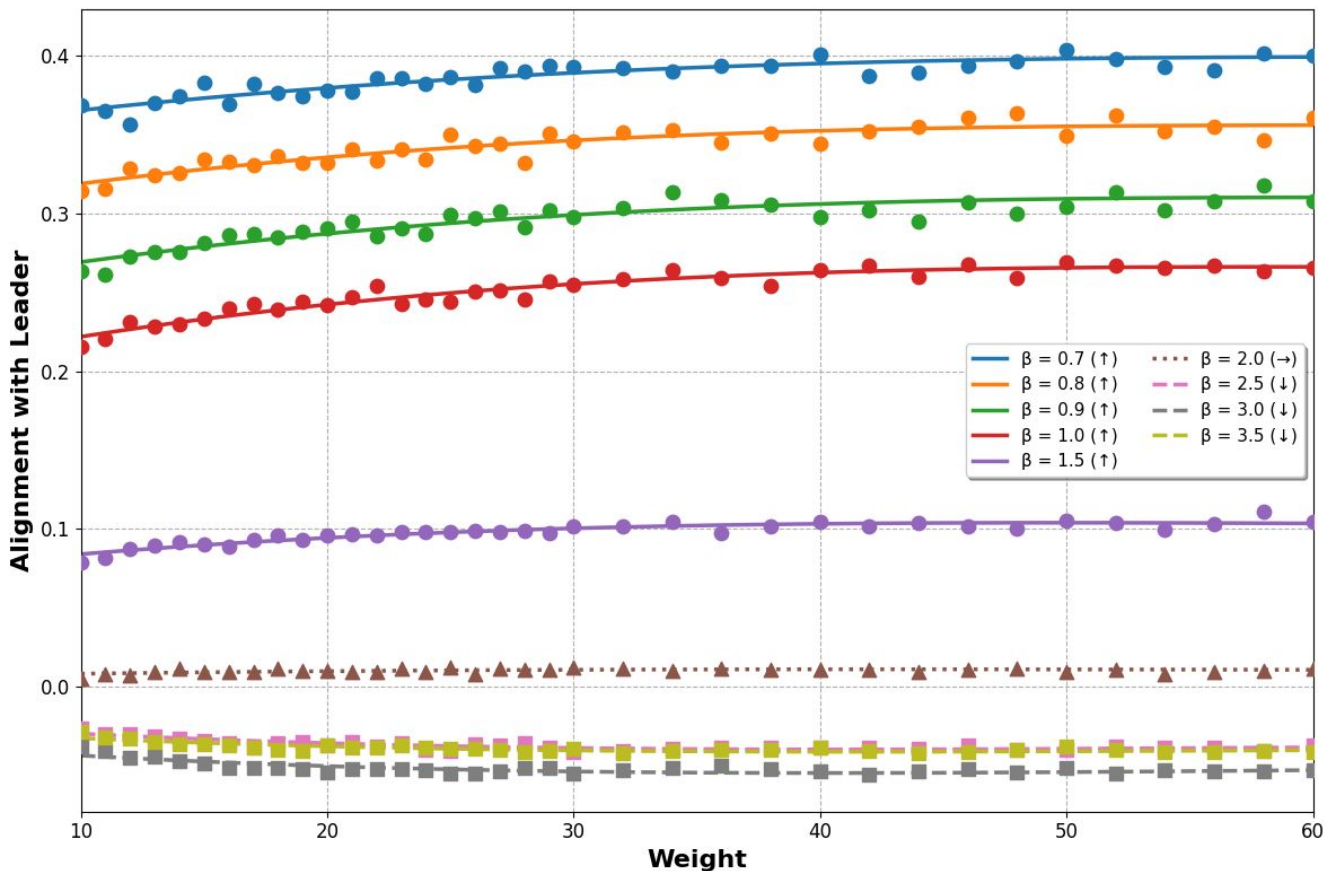
$$\theta_i(t + \Delta t) = \frac{1}{w} \langle \theta_j(t) \rangle_{j \in \mathcal{N}_i \setminus \{i\}} + \theta_i(t) + \beta \xi_i(t)$$



A strong but unreliable signal being injected into the flock.

Mean Alignment vs. weight

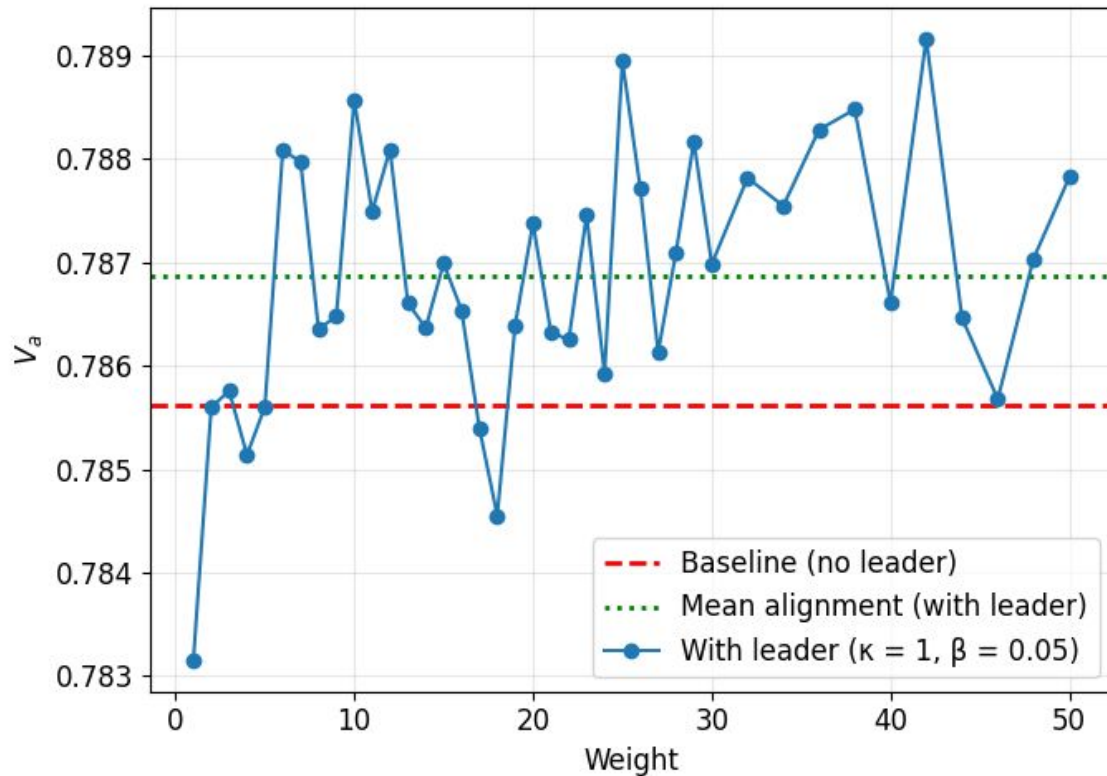
$$\theta_i(t + \Delta t) = \frac{1}{w} \langle \theta_j(t) \rangle_{j \in \mathcal{N}_i \setminus \{i\}} + \theta_i(t) + \beta \xi_i(t)$$



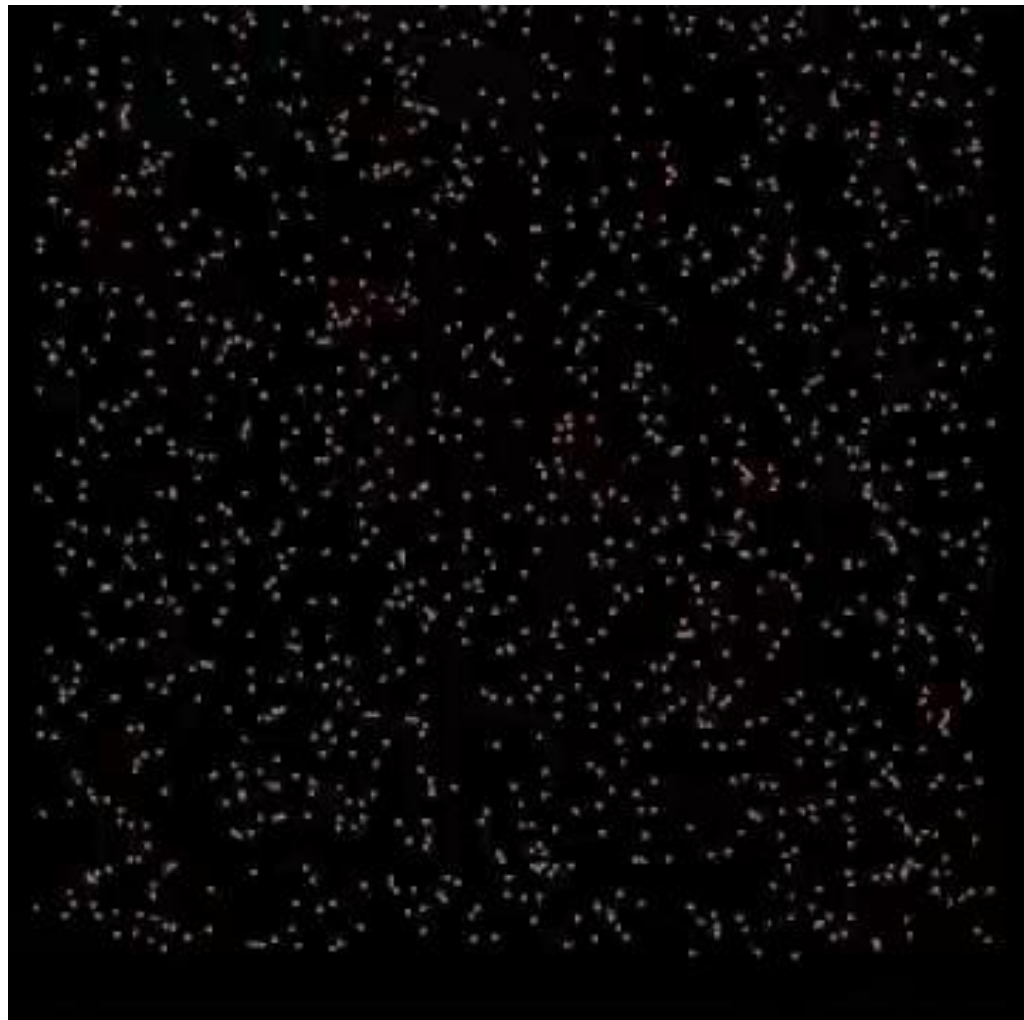
- **Small β** : helps alignment
- **Medium β** : Natural
- **Large β** : kills alignment

Potential Scenario

Leaders completely coupled with the neighbors, but know where to go.



- κ is how much the bias affect the leader.





Thank you for your attention :)